Topic 4 — Materials

4.1 Density

Density (ρ): The mass per unit volume of a material. It is a scalar property of the material itself (not the object). Units: kg m^{-3}.

$$\rho = m / V$$

Measuring Density

- **Regular solids:** Measure mass with a balance. Calculate volume from dimensions (cuboid: $V = l \times w \times h$; cylinder: $V = \pi r^2 h$; sphere: $V = (4/3)\pi r^3$).
- **Irregular solids:** Measure mass. Find volume by the displacement method: submerge the object in water in a measuring cylinder and record the rise in water level ($V = V_{final} - V_{initial}$). Alternatively use a eureka can (displacement can).
- **Liquids:** Measure the mass of a known volume using a measuring cylinder and balance. Subtract the mass of the empty cylinder.

Material	Density (kg m^{-3})
Air (at STP)	1.2
Water	1000
Aluminium	2700
Iron/Steel	7800

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Copper	8900
Lead	11300
Gold	19300

Worked Example: A metal cylinder has diameter 2.4 cm, height 5.0 cm, and mass 180 g. Calculate its density and identify the likely metal.

Solution:
r = 1.2 cm = 0.012 m; h = 0.050 m; m = 0.180 kg
$V = \pi r^2 h = \pi \times (0.012)^2 \times 0.050 = \pi \times 7.2 \times 10^{-6} = 2.262 \times 10^{-5}\ m^3$
$\rho = m/V = 0.180 / 2.262 \times 10^{-5} =$ **$7960\ kg\ m^{-3}$**
This is close to the density of steel ($7800\ kg\ m^{-3}$) or iron ($7870\ kg\ m^{-3}$).

Exam Tip: An object floats in a fluid if its average density is less than the fluid's density. An object sinks if its average density is greater. This is why steel ships float — the hull encloses a large volume of air, so the average density of the ship is less than water.

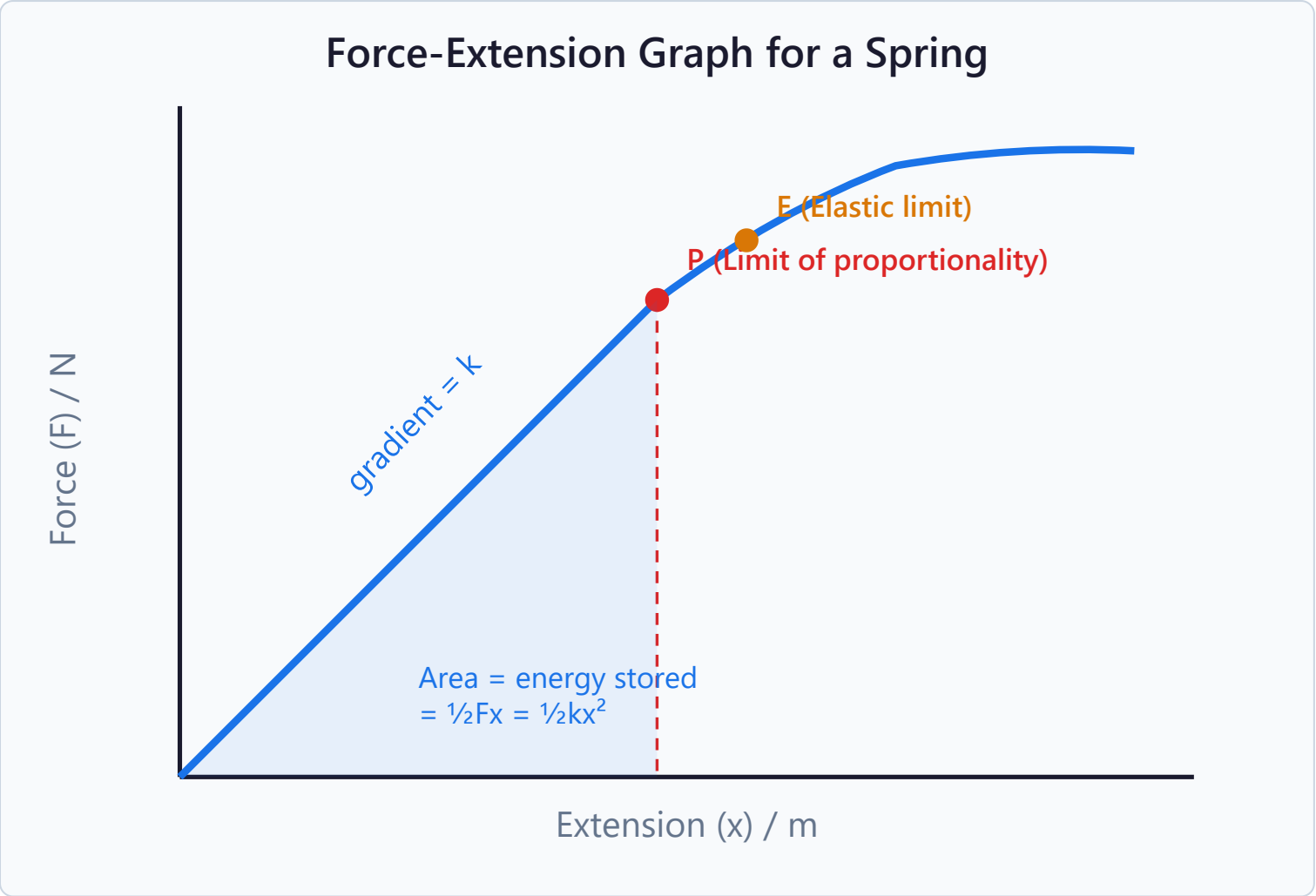
4.2 Hooke's Law

Hooke's Law: The extension of a spring (or other elastic object) is directly proportional to the applied force, provided the limit of proportionality is not exceeded.

$$F = kx$$

where F = applied force (N), k = spring constant ($N\ m^{-1}$), x = extension (m).

The **spring constant (k)** is the force per unit extension — a measure of stiffness. A stiffer spring has a larger k. It equals the gradient of the linear section of a force-extension graph.



Key Points on the Force-Extension Graph

- **Limit of proportionality (P):** Up to this point, $F \propto x$ (Hooke's law holds). Beyond P, the graph is no longer linear, but the material may still return to its original length.
- **Elastic limit (E):** Up to this point, the material returns to its original length when the force is removed (elastic behaviour). Beyond E, permanent (plastic) deformation occurs. E is at or slightly beyond P.
- **Yield point:** The point where the material begins to deform plastically with little or no increase in force (the material "gives" or "flows").

Springs in Series and Parallel

Arrangement	Combined Spring Constant	Effect
Series (end to end)	$1/k_{total} = 1/k_1 + 1/k_2$	k_{total} is LESS than either individual k (softer overall)

Parallel (side by side)	$k_{total} = k_1 + k_2$	k_{total} is MORE than either individual k (stiffer overall)

Worked Example: Two springs each have k = 40 N m^{-1}. Find the combined spring constant when they are arranged (a) in parallel, (b) in series. A 2.0 N force is applied. Find the extension in each case.

Solution:

(a) Parallel: k_{total} = 40 + 40 = **80 N m^{-1}**. Extension x = F/k = 2.0/80 = **0.025 m** (25 mm)

(b) Series: $1/k_{total}$ = 1/40 + 1/40 = 2/40 = 1/20, so k_{total} = **20 N m^{-1}**. Extension x = 2.0/20 = **0.10 m** (100 mm)

The series arrangement extends four times more than the parallel arrangement for the same force.

Exam Tip: Springs in series combine like resistors in parallel (reciprocal rule — total k decreases). Springs in parallel combine like resistors in series (add directly — total k increases). The physical intuition: in parallel, the load is shared between the springs, so each extends less; in series, each spring carries the full load.

4.3 Elastic Potential Energy

Elastic potential energy (E_e): The energy stored in a stretched or compressed elastic object. It equals the work done in deforming the object (the area under the force-extension graph in the linear region).

$$E_e = \frac{1}{2}Fx = \frac{1}{2}kx^2$$

This equation only applies within the limit of proportionality (where F = kx). The factor of ½ arises because the force increases linearly from zero to F as the spring is stretched, so the average force is F/2.

For a non-linear force-extension graph, the energy stored equals the area under the curve, which must be estimated (e.g. by counting squares or using the trapezium rule).

Worked Example: A spring with k = 50 N m^{-1} is compressed by 0.080 m. Calculate the elastic PE stored. If the spring launches a 0.020 kg ball vertically, find the maximum height reached (assuming no energy losses).

Solution:

$E_e = ½kx^2 = ½ \times 50 \times (0.080)^2 = ½ \times 50 \times 0.0064 =$ **0.16 J**

At max height: $E_e = mgh$

$0.16 = 0.020 \times 9.81 \times h$

$h = 0.16 / (0.020 \times 9.81) =$ **0.82 m**

4.4 Stress, Strain, and Young's Modulus

Tensile stress (σ): The force per unit cross-sectional area acting on a material. Units: Pa (N m^{-2}).
Tensile strain (ε): The fractional change in length. It is dimensionless (no units).
Young's modulus (E): The ratio of stress to strain in the linear elastic region. It measures the *stiffness* of a material. Units: Pa.

$$\text{Stress: } \sigma = F / A$$

$$\text{Strain: } \varepsilon = \Delta L / L$$

$$\text{Young's modulus: } E = \sigma / \varepsilon = FL / (A\Delta L)$$

Why use stress and strain instead of force and extension? Because stress and strain are **material properties** — they allow us to compare different materials regardless of the sample's dimensions. A short thick wire and a long thin wire of the same material have the same Young's modulus, even though they have very different force-extension characteristics.

Material	Young's Modulus (GPa)	UTS (MPa)
Rubber	~0.01–0.1	~15

Bone	~15	~130
Copper	~120	~220
Steel	~200	~400–800
Diamond	~1200	—

Worked Example: A steel cable of diameter 5.0 mm and length 20 m supports a mass of 500 kg. Calculate (a) the stress in the cable, (b) the strain, (c) the extension. (E_{steel} = 200 GPa)

Solution:

$A = \pi(d/2)^2 = \pi \times (2.5 \times 10^{-3})^2 = 1.963 \times 10^{-5}\ m^2$

$F = mg = 500 \times 9.81 = 4905$ N

(a) $\sigma = F/A = 4905 / 1.963 \times 10^{-5} = \mathbf{2.50 \times 10^8\ Pa} = 250$ MPa

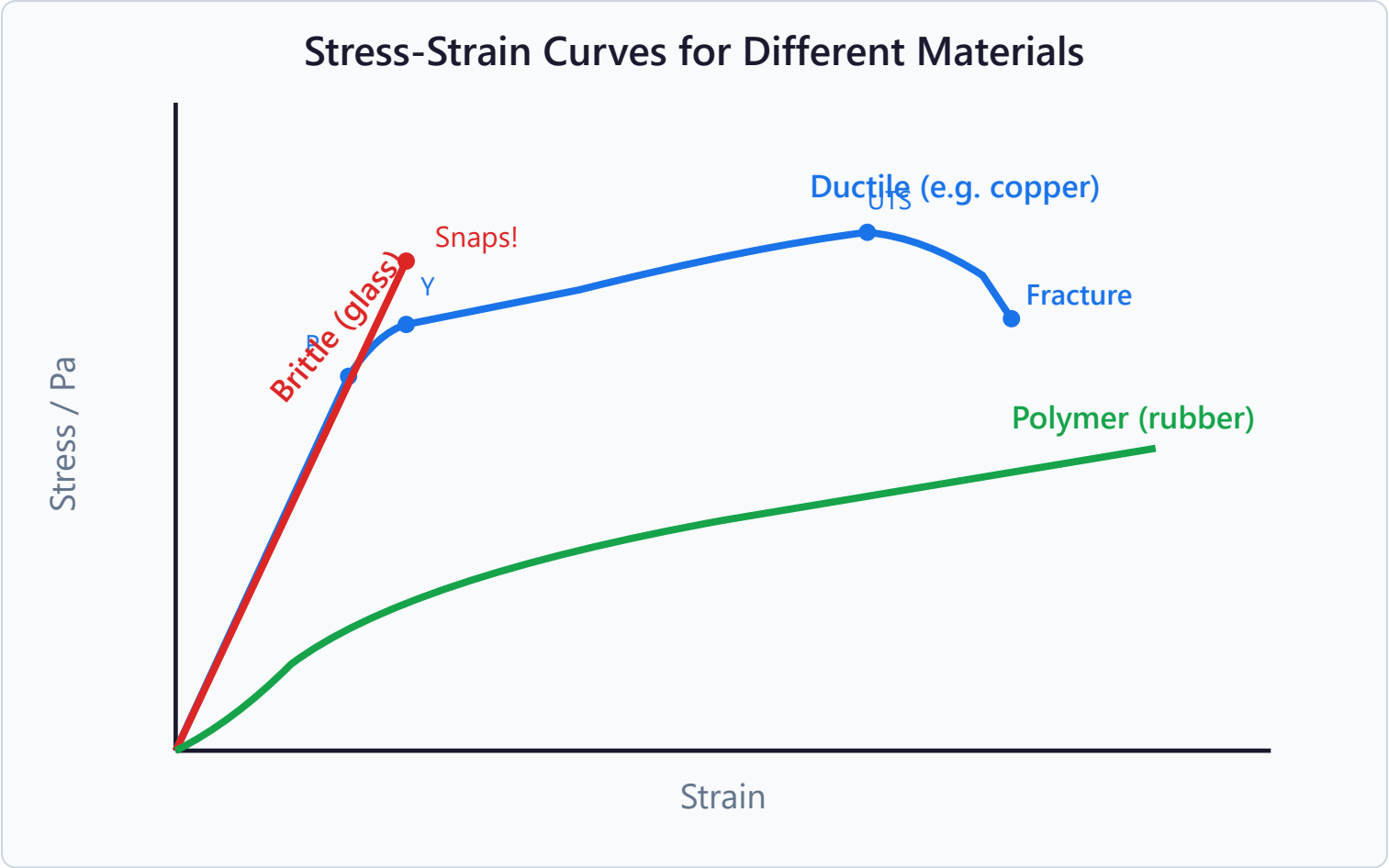
(b) $\varepsilon = \sigma/E = 2.50 \times 10^8 / 200 \times 10^9 = \mathbf{1.25 \times 10^{-3}}$ (0.125%)

(c) $\Delta L = \varepsilon \times L = 1.25 \times 10^{-3} \times 20 = \mathbf{0.025\ m}$ (25 mm)

4.5 Stress-Strain Graphs

Key Points on a Stress-Strain Graph (Ductile Material)

- **P (Limit of proportionality):** End of the linear region; beyond this, stress is no longer proportional to strain.
- **E (Elastic limit):** Beyond this, permanent plastic deformation occurs. The material will not return to its original length.
- **Y (Yield point):** A large increase in strain occurs with little increase in stress. The material begins to "flow."
- **UTS (Ultimate tensile stress):** The maximum stress the material can withstand. Beyond this, the material "necks" (cross-section decreases locally).
- **F (Fracture point):** The material breaks.



4.6 Material Properties

Property	Definition	Examples
Stiff	High Young's modulus; resists elastic deformation (steep initial gradient on stress-strain graph)	Steel, diamond, bone
Strong	High UTS; withstands large stresses before breaking	Steel, kevlar, carbon fibre
Tough	Absorbs a lot of energy before fracturing (large area under stress-strain curve)	Mild steel, rubber
Ductile	Can be drawn into wires; large plastic deformation before fracture	Copper, gold, mild steel

Brittle	Fractures with little or no plastic deformation; snaps suddenly	Glass, ceramics, cast iron
Hard	Resistant to scratching or surface indentation	Diamond, tungsten carbide
Malleable	Can be hammered or pressed into shape	Gold, aluminium, copper

Exam Tip: Do not confuse stiff with strong. A material can be stiff but weak (e.g. glass — high Young's modulus but low UTS), or flexible but strong (e.g. nylon rope). "Stiff" = resists deformation (high E). "Strong" = resists breaking (high UTS). "Tough" = absorbs energy before breaking (large area under the graph).

4.7 Elastic and Plastic Deformation

Elastic deformation: The material returns to its original shape and size when the force is removed. No permanent change. Energy is fully recoverable. Occurs below the elastic limit.

Plastic deformation: The material is permanently deformed and does not return to its original dimensions when the force is removed. The energy used to cause plastic deformation is dissipated as heat (not recoverable).

Hysteresis

When rubber is loaded and then unloaded, the unloading curve lies below the loading curve on a force-extension graph. The area enclosed between the two curves represents the **energy dissipated as thermal energy** during one loading-unloading cycle. This is called **hysteresis**.

Rubber returns to its original length (elastic), but the energy "lost" to hysteresis heats the rubber. This is why repeatedly stretching a rubber band makes it warm.

4.8 Energy Stored in Deformation

- The **area under a force-extension graph** = energy stored (in joules).
- The **area under a stress-strain graph** = energy stored per unit volume (in J m^{-3}).
- For a material obeying Hooke's law: Energy = $\frac{1}{2}Fx = \frac{1}{2}kx^2 = F^2/(2k)$.
- If loading and unloading curves differ (hysteresis), the energy returned on unloading = area under the unloading curve. The energy dissipated = area between the curves.

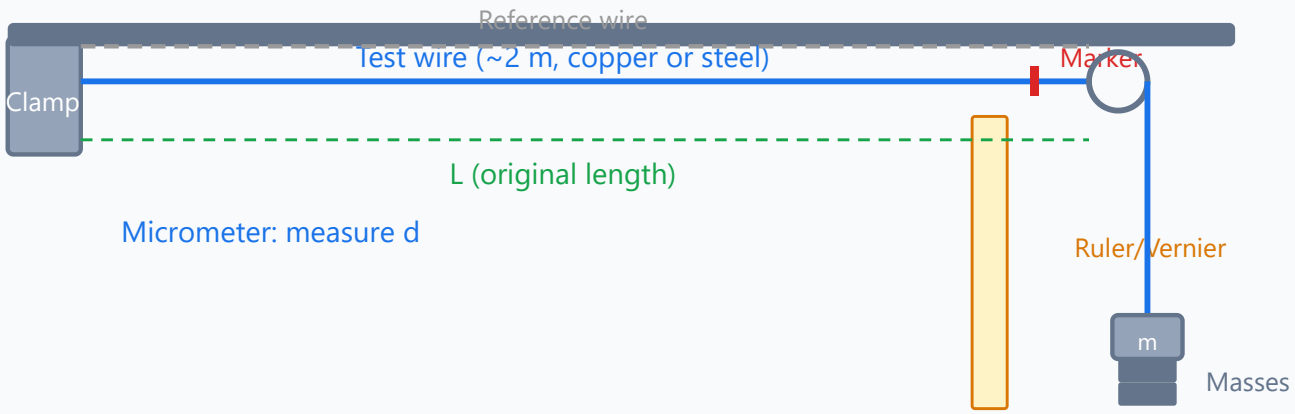
Exam Tip: Do not confuse force-extension graphs (energy in J) with stress-strain graphs (energy per unit volume in J m^{-3}). The area under a stress-strain curve gives energy per unit volume. To get total energy, multiply by the volume of the sample.

Summary of Key Equations

Quantity	Equation	Units
Density	$\rho = m/V$	kg m^{-3}
Hooke's Law	$F = kx$	N, N m^{-1}, m
Elastic PE	$E = \frac{1}{2}kx^2 = \frac{1}{2}Fx$	J
Stress	$\sigma = F/A$	Pa (N m^{-2})
Strain	$\varepsilon = \Delta L/L$	dimensionless
Young's modulus	$E = \sigma/\varepsilon = FL/(A\Delta L)$	Pa
Springs in series	$1/k_T = 1/k_1 + 1/k_2$	N m^{-1}
Springs in parallel	$k_T = k_1 + k_2$	N m^{-1}

Core Practical 3: Determine the Young modulus of a material

Young's Modulus: Wire Experiment Setup



E = stress / strain = gradient of stress-strain graph

Aim

To determine the Young modulus of a metal wire by measuring stress and strain.

Apparatus

Long thin wire (~2 m, copper or steel), reference wire, micrometer (0.01 mm), metre ruler or tape, vernier scale or travelling microscope, mass hanger and slotted masses, safety goggles, G-clamp.

Method

1. Clamp both wires to a rigid beam. Attach a small mass to the reference wire to keep it taut.
2. Measure the original length L of the test wire with a tape measure.
3. Measure the wire diameter d with a micrometer at 3+ positions and orientations. Calculate mean d and area $A = \pi(d/2)^2$.
4. Zero the vernier scale. Add masses in steps (e.g. 100 g increments). Read the extension ΔL after each addition.
5. Record at least 6 loads. Do not exceed the elastic limit (check by removing the final mass).
6. Calculate stress = F/A and strain = ΔL/L for each reading.
7. Plot stress vs strain. Gradient of the linear region = Young's modulus E.

Safety

Wear safety goggles (wire may snap). Place a cushion beneath masses. Do not stand below the loaded wire.

Sources of Error & Improvements

Source of Error	Type	Improvement
Wire diameter varies (not perfectly circular)	Random	Measure at 6+ positions and orientations; use mean
Wire has kinks (not straight)	Systematic	Apply a small initial load to straighten before zeroing
Small extensions hard to measure	Random	Use a long, thin wire to maximise extension; use travelling microscope
Thermal expansion changes length	Systematic	Use a reference wire — both expand equally, so the vernier reading is unaffected
Wire creeps under load	Systematic	Do not exceed elastic limit; check by unloading periodically

Exam Tip: The reference wire compensates for thermal expansion. Both wires expand equally with temperature, so the vernier reading (measuring the difference in extension) is unaffected. This is the key reason for using a reference wire — always state this in your answer. The biggest uncertainty is usually in the diameter measurement (squared in the area formula, so uncertainty is doubled).